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The effect of planting date on maize grain yields and yield components

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ABSTRACT

Four experiments were established in the Waikato and Manawatu regions of New Zealand in 2006 and 2007 to determine planting date (PD) effect on maize (*Zea mays* L.) leaf growth, grain yield (GY) and yield components. Five or six hybrids of three maturity classes (early, mid and late) were sown on four or five PDs between 18 September and 15 December of each year. Increasing mean daily temperatures in the range 13–19 °C immediately prior to tassel initiation reduced leaf number by 0.1 leaf °C^{-1}. Highest leaf area indices were observed at mean daily temperatures of 17–19 °C. In the lower latitude environment of Waikato, maximum GY was obtained with earlier plantings than Manawatu. Lower spring temperatures, and consequently smaller canopy sizes in Manawatu depressed yields of early plantings. When planted early, late hybrids generally outyielded early hybrids while a better balanced source–sink ratio meant that early hybrids yielded consistently across PDs, matching or outyielding late hybrids when both were planted late. Lower grain filling mean temperatures (15 vs. 18 °C) and average radiation (11 vs. 20 MJ m^{-2} d^{-1}) reduced yields more for late than early plantings. Grain yield was highly correlated with kernel number (KN) ($r = 0.90^{***}$) and weight (KW) ($r = 0.76^{***}$). Lowest KN, KW and GY values were obtained under late plantings, low rainfall (<20 mm) and/or radiation (<18 MJ m^{-2} d^{-1}) 10–20 d either side of flowering, or when mean temperatures ≤15 °C or irradiance <11 MJ m^{-2} d^{-1} occurred during grain filling. Kernel number, KW and GY responses to late planting or water stress were more apparent in late than early hybrids. Kernel weight was more stable than KN under late planting or water stress conditions. Water stress during grain filling affected late PDs more than early PDs. Total biomass and harvest index decreased with delayed planting.

1. Introduction

Environmental factors such as temperature, radiation and water vary in a random manner around their seasonal mean trends, resulting in uncertainties in crop production during the main cropping season. The biggest challenge is to determine the ideal time for planting either early or late season hybrids, especially when planting window is shortened by weather or management challenges. Because of differences in maturity and length of growing seasons, the ideal planting dates (PDs) for maize (*Zea mays* L.) hybrids vary among locations and seasonally within locations. To maximise yields and profitability it is however essential that the maize hybrid with the right maturity be selected for planting given the planting window available to the farmer.

Research has shown a yield penalty from planting early hybrids if season length was sufficient for later maturing hybrids (Sorensen et al., 2000). Early maturing hybrids usually fail to fully utilise available solar radiation for the period when temperatures are suitable for growth and therefore will not realise the full yield potential of the growing season and the inputs provided by the grower (Lauer, 1998). Similarly, late hybrids may fail to mature before the first killing frost occurs. When planted late, early hybrids could equal, or outperform full season hybrids (Staggenborg et al., 1999). Early hybrids may also be more profitable since later hybrids may require additional artificial drying for safe storage.

Environment by crop growth or development interactions influence crop cycle duration and yields differently. For instance, if intercepted photosynthetically active radiation (IPAR) levels per plant around flowering are low, kernel set declines and harvest index (HI) may be significantly reduced. Under favourable growing conditions, assimilate production rate, which determines maize yields, is driven by IPAR and leaf photosynthetic rate (Monteith, 1977). Radiation interception is largely determined by leaf area

Abbreviations: ASI, anthesis-silking interval; ENV, environment; GY, grain yield; HI, harvest index; IPAR, intercepted photosynthetically active radiation; KN, kernel number; KW, kernel weight; LA, leaf area; LAI, leaf area index; NIWA, National Institute of Water and Atmospheric Research; PD, planting date; PM, physiological maturity; RUE, radiation use efficiency; TI, tassel initiation.

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index (LAI) which in turn is influenced by genotype, nitrogen and water supply, temperature and photoperiod (Muchow and Carberry, 1989).

High grain yields (GYs) result from an increased availability of assimilate supply (source) for grain setting and grain filling accompanied by an enhanced capacity of the kernels (sink) to accommodate those assimilates. Source–sink balance alters when crops are planted either early or late. Kernel number is usually strongly associated with GY (Otegui et al., 1995; Bolaños and Edmeades, 1996) and therefore defines sink size. Kernel numbers per plant are largely determined by conditions that occur around silking time (Andrade et al., 1993; Kiniry and Knievel, 1995) and therefore have a large bearing on GY. Low temperatures encountered with early planting tend to reduce plant height (Al-Darby and Lowery, 1987) mainly by decreasing internode length and less so by reducing leaf numbers. Leaf area (LA) is also considerably lower.

Yield reductions due to early or late planting are well documented in the literature (e.g. Johnson and Mulvaney, 1980; Sorensen et al., 2000). While early planting results in reduced IPAR because of delayed LA development, high temperatures under late planting situations also reduce IPAR in the two week period either side of flowering by reducing calendar time for crop development, thereby decreasing yields (Otegui et al., 1996). Cool nights during grain filling, common under late planting situations, may also reduce radiation use efficiency (RUE) (Jones et al., 1986).

In situations where mid-season drought is likely to occur, late planting can also expose crops to water deficits during flowering. This may delay silking, increasing the anthesis-silking interval (ASI) resulting in reduced sink size through poorly pollinated ears, or through abortion of kernels and ears (Bolaños and Edmeades, 1996). Anthesis-silking interval is a predictor of seed set in many maize cultivars when under stress at flowering (Edmeades et al., 2000). Drought effects can thus be minimised by adjusting planting time and hybrid maturity so that flowering occurs when drought risk is judged to be minimal.

Where planting has been delayed, or where hybrids are being sown in new and untested areas, knowledge of how the maturity of the hybrid interacts with its response to variable environmental conditions is key to developing mitigating strategies required to optimise and stabilise yields. If these relationships are generally known, they can be used to develop response functions to PD by hybrid, and craft a strategy that will maximise farmers' incomes.

The objective of this study was to determine how PD and the environment (ENV) affect harvest index (HI), sink and source size, barrenness and consequently grain and total biomass yields of maize hybrids differing in maturity in a cool temperate climate. The study is part of a more extensive investigation that forms the basis of adapting the CERES Maize model to predict an optimal match of hybrid and environment when planting maize in a temperate climate (Tsimba, 2011).

2. Materials and methods

2.1. Site details

Four replicated experiments were established over two growing seasons at three locations in New Zealand. These were at Rukuhia Research Station (37.86°S, 175.32°E; 50 m asl) in 2006 (RUK07) and 2007 (RUK08), Ngaroto Research Station (37.98°S, 175.32°E; 84 m asl) (NGA08) and Massey University Pasture and Crop Research Unit (40.38°S, 175.58°E; 18 m asl) in 2007 (MAS08). The latter was situated in the Manawatu Region on a Manawatu fine sandy loam (Dystric Fluventric Eutrochrept). Both Rukuhia and Ngaroto Research Stations are situated in the Waikato Region on a Horotiu sandy loam soil (Vitric Orthic Allophanic) and an Ohaupo silt loam (Typic Orthic Allophanic), respectively (Hewitt, 1998).

The Manawatu site had a history of long term pasture whereas both Waikato sites had been in long term maize monoculture for up to 25 yr. The history of the Waikato sites closely mirrors a typical farming system common in New Zealand for maize grain production, while Manawatu more closely characterises silage production under a dairy farming system. The selected sites can also be considered to be a fair representation of soil and weather conditions for their respective regions. Rukuhia and Ngaroto respectively typify low and high potential yielding sites for their regions, and differ mainly in soil type. Because of its higher latitude, the Manawatu site experiences a shorter growing season with cooler mean spring temperatures and a more rapid decline in solar radiation and temperature during the autumn, compared to the Waikato sites.

2.2. Planting details and experimental design

Five and four PD treatments were established at the Waikato ENVs and MAS08, respectively. The range of PDs and ENVs evaluated in this study were designed to create contrasting environmental conditions that represent a wide range of situations for maize growth and development. Planting dates were thus widely spread to include very early, typical and very late PDs for each ENV, and were as follows:

RUK07 – 18 September, 11 October, 2 November, 24 November, and 15 December, 2006.
RUK08 – 20 September, 13 October, 1 November, 22 November, and 13 December, 2007.
NGA08 – 21 September, 11 October, 1 November, 22 November, and 13 December, 2007.
MAS08 – 16 September, 6 November, 23 November, and 10 December, 2007.

Hereafter, PD treatments will be referred to as PD1, PD2, PD3, PD4 and PD5, respectively. At MAS08, PD1 was considered missing in subsequent data analyses.

Six single cross hybrids representing three maturity groups of comparative relative maturity (CRM) ratings of 91–110; short season (38P05 and 38H20), mid season (36M28 and 36B08) and full season (34D71 and 34P88) were planted in the Waikato ENVs. Hereafter, hybrid maturities will be referred to as early, mid and late, respectively. Hybrid maturity designations varied with ENV. At MAS08, 36M28, 36B08 (late), 38H20, 38P05 (mid) and 39G12 (early) were planted. The CRM ratings from hybrids planted at MAS08 ranged between 78 and 103. The later maturity hybrids sown in the Waikato were considered too late and not adapted to the Manawatu Region, whereas 39G12 was regarded as too early for the Waikato ENVs. These hybrids were selected from a pool of 29 Pioneer® hybrids that are commercially available in New Zealand, and are representative elite hybrids for their maturity classes.

Prior to planting, soil tests were conducted on each ENV to determine the nutrient level of each field. Soil pH, Olsen P, cation exchange capacity (CEC), base saturation, sulphate-S, total C and N and organic matter in the upper 60 cm were determined in order to calculate fertiliser and lime requirements needed to ensure non-limiting nutrient supply (Edmeades et al., 1984). Soil water contents and bulk density for each soil horizon within the profile for each PD treatment were measured gravimetrically at or prior to planting. The soil samples were obtained from three horizon layers (0–20 cm, 20–45 cm and 45–65 cm) using soil cores of 4.8 cm diameter by 5 cm height, taken from the midpoint of each horizon. The samples were weighed immediately and oven dried to constant weight at 105 °C.

In early spring, the first experiment (RUK07) received 92, 50 and 50 kg ha^{-1} of N, P and K, respectively, as base fertiliser in the form of Cropzeal (12:10:10) and urea (46-0-0). Soon after base fertiliser application, the trial area was disc ploughed to 25 cm, and immediately before planting, the seed bed was cultivated to remove weeds and to loosen the soil. At planting, Nitrophoska (12-10-10) was applied as starter fertiliser at a rate of 400 kg product ha^{-1}. When six fully expanded leaves had emerged, 365 kg ha^{-1} urea was applied as a sidedressing. In total, both NGA08 and RUK08 received 324 kg N, 81 kg P and 40 kg K ha^{-1}. Based on initial soil test results, MAS08 did not require base or starter fertiliser, but received a total of 220 kg N ha^{-1} applied as a sidedressing of urea.

Weeds were controlled by a combination of 3 l ha^{-1} Roustabout® (840 g a.i. l^{-1} acetochlor) and 3 l ha^{-1} atrazine (600 g a.i. l^{-1} atrazine) applied as a pre-emergent herbicide at planting. Additional weed control was obtained by applying 200 ml ha^{-1} Callisto® (480 g a.i. l^{-1} mesotrione) in combination with 1.5 l ha^{-1} Gesaprim® (500 g a.i. l^{-1} atrazine) as post emergence herbicides prior to canopy closure.

Each experiment was designed as a randomised complete block design with a split-plot treatment arrangement replicated three times. Planting dates were considered as the main plots and hybrids as sub-plots, each randomised within the next higher treatment level. Planting dates were randomised within the field to avoid confounding PD effects with trends in soil fertility or texture.

Apart from PD and hybrid treatments, all other management factors were held constant. To minimise impact of adjacent treatments in six row plots, the two outside rows in each plot were considered as borders. Plot length details are discussed in Section 2.6. Approximately 1 m at either end of the harvested or sampled plot area was also considered as border. Planting date treatments were separated by an additional four to six border rows on either side of each treatment. The Waikato experiments were planted using a 2-row Wintersteiger® precision vacuum seeder while the MAS08 experiment was planted using a modified 2-row cone seeder.

To achieve uniform plant density, all plots were planted at 130,000 plants ha^{-1} and thinned at the V4 leaf stage to achieve population densities of 110,000 plants ha^{-1} for 39G12, 38H20 and 38P05 and 105,000 plants ha^{-1} for the rest. These densities were based on agronomic planting density optima required to maximise both grain and silage yields under unstressed conditions (Genetic Technologies Ltd., 2004, *unpublished data*).

2.3. Weather

Daily weather data (rainfall, air temperature, solar radiation) between planting and harvesting were obtained from the nearest National Institute of Water and Atmospheric Research (NIWA) automated weather stations, ranging in distance from 400 m to 12 km from the plots. Temperatures at the level of the apical meristem were recorded for each PD treatment every hour using "WatchDog 100" (Spectrum Technologies®, Inc.) data loggers wrapped in plastic film and inserted in the soil at approximately 5 cm depth between rows in the trial plot area. Seven irrigations totalling 178 mm water were applied by overhead sprinklers at MAS08 to prevent the crop suffering from any visual symptoms of drought stress. Irrigation was not available at the Waikato sites.

2.4. Photoperiod sensitivity

Photoperiod was estimated for a period 3–5 weeks after planting using the Martindale Centre Photoperiod Calculator (Lammi, 2008). Effective photoperiod is usually estimated as the time between the start and end of successive civil twilight periods spanning a period of daylight.

2.5. Leaf number and leaf area index

Before the first leaf had senesced, 10 plants in the centre two rows (five consecutive plants per row) were tagged by cutting the tip of leaf five, considering the leaf with the characteristic rounded tip as leaf one. Leaf 10 was also tagged as soon as it appeared. Tagged leaves were used as reference points for convenient and accurate counting of leaf number after the lower leaves had senesced.

At anthesis, four consecutive plants were selected for green LA measurements in each plot. The length (from ligule to leaf tip) and width (the widest portion of the leaf blade) of each green leaf were measured in situ. Individual green LA was estimated as the product of leaf length and maximum breadth, adjusted by a constant coefficient as follows: $LA = length \times maximum\ width \times 0.75$ (Elings, 2000). Leaf area per plant was calculated by summing the individual LAs. Leaf area index was calculated as the average plant LA divided by the average land area occupied by a single plant.

2.6. Grain yield, grain yield components and total biomass

At RUK07, a bordered area of 7.6 m^2 (2.5 m × 0.76 m × 4 rows) in each plot was hand-harvested for GY at least 2 wk after physiological maturity (PM). Harvested areas for the other ENVs were 9.8 m^2 (3.5 m × 0.7 m × 4 rows), 5.9 m^2 (3.9 m × 0.76 m × 2 rows) and 11.9 m^2 (3.9 m × 0.76 m × 4 rows) for MAS08, NGA08 and RUK08, respectively. Different harvest areas were a result of variable paddock sizes available for the trial work.

Harvested ears were counted, and ears containing <10 kernels were considered barren. Ears were then shelled with a portable sheller, and grain tested for moisture using a GAC 2100 grain analysis metre (DICKEY-John®, Auburn, IL). Grain yield was adjusted to a 14% moisture content basis. At RUK07, frost damage occurred before late hybrids in PD5 had matured, resulting in the grain not being harvested. This was consequently treated as a missing data point.

Individual kernel weight (KW) per plot was estimated from a random subsample of 1000 kernels. Number of kernels (KN) m^{-2} was estimated as: total plot grain weight m^{-2}/KW. Average KN $plant^{-1}$ was estimated as the ratio of KN m^{-2} and plants m^{-2}. Sink size was assumed to be equivalent to GY $plant^{-1}$ (or KN × KW (g)). To estimate HI, stover from the plot areas harvested for grain was cut at ground level, immediately weighed, and a known sub sample dried to constant weight at 75 °C.

2.7. Data analysis

Analyses of variance (ANOVA) to determine effects of PD on maize hybrid performance and its interactions with hybrid maturity were conducted using the Mixed Model Procedure (Proc Mixed) in SAS (Littell et al., 2006). Hybrid within a maturity group was considered a random factor, while all other factors were treated as fixed effects. Where appropriate, estimate and contrast statements in SAS were used to estimate and compare the linear combination of PD and hybrid maturity main effects. Combined analyses across ENVs were conducted to determine ENV × treatment interaction effects. The General Linear Model Procedure of SAS (Proc GLM) was used to compute the regression of measured traits on PD, while simple correlations (Proc CORR) were used to determine the nature of relationships among the measured traits. Where correlations among traits were similar across ENVs and/or hybrid maturities, correlation data were aggregated across ENVs and maturity classes.

All data analyses across ENVs resulted in significant ENV × maturity × PD, ENV × maturity or ENV × PD interactions, suggesting that planting or maturity effects were ENV dependent. Due to a significant lack of homogeneity of variance across the four ENVs, data from the four experiments (i.e., years and locations) were treated as independent ENVs and analysed separately (Gomez and Gomez, 1984; Milliken and Johnson, 1994). Although the Rukuhia experiment was repeated in successive years, the two seasons were quite different, with the first being typical and the second being extremely dry. The first season therefore represented a soil ENV with a relatively low yield potential under normal rainfall conditions, whereas RUK08 was characteristic of that same soil in a very dry season.

For data presentation, the following notations *, ** and *** are used to indicate significance at $0.01 < P < 0.05$, $0.001 < P < 0.01$, and $P < 0.001$ respectively, while NS refers to $P \geq 0.05$.

3. Results

3.1. Weather data

Tables 1 and 2 illustrate the average monthly weather data during the experimental period for the four ENVs (RUK07, RUK08, NGA08 and MAS08). NGA08 and RUK08 experienced temperatures in the dry hot summer of 2008 (27 ± 0.4 °C) that were 3 °C higher than their long term averages (NIWA, 2001). Thereafter, temperatures were about 1 °C higher than average, before reaching the long term average in April. The T_{max} values for MAS08 were 1–2 °C warmer than average between January and March while RUK07 temperatures resembled long term averages for the region.

Rainfall was moderate at RUK07, while drought stress effects were visually evident at the other three ENVs. Unlike MAS08 where irrigation was applied, water stress was severe during the late December to late March period for RUK08 and to a lesser extent for NGA08. RUK07 was the only ENV to have experienced at least some precipitation every 2 wk during the growing season. Between 1 January and 31 March (critical flowering and grain setting period covering all PD treatments) the total amounts of precipitation were 260, 150, 59 and 61 mm for RUK07, MAS08, NGA08 and RUK08, respectively, while cumulative Penman PET amounts were 336, 410, 386 and 381 mm for the same periods.

Across all ENVs, solar radiation was greatest between early November and late February, with the highest readings recorded between 22 December and 5 January. RUK07 experienced the least daily average solar radiation during the critical grain filling period of January to March (19 MJ m^{-2} d^{-1}) compared to ≥20 MJ m^{-2} d^{-1} for the other ENVs. A significant reduction in solar radiation in April was evident under all ENVs, with values falling to about 50% of January levels.

Despite the higher radiation levels during early developmental stages for late plantings, early sown crops generally accumulated more total radiation between emergence and flowering due to slower plant development under the lower temperature conditions (Table 3). At RUK07, for example, average planting to flowering duration for PD1 was 108 d vs. 67 d for PD5. A general steady increase in mean temperature from PD1 to 5 characterised all ENVs. There was comparatively little variation in daylengths estimated for the period one week before tassel initiation (≤0.6 h) among PDs in all ENVs (Tsimba et al., 2013).

3.2. Leaf number and leaf area index

Across hybrid maturity classes the average total leaf number was highest at NGA08 (18.7) and RUK08 (18.6). RUK07 averaged 18.4 leaves across treatments while MAS08, which had an earlier set of hybrids, had the lowest total leaf number (17.9).

At NGA08, PD had a significant effect on leaf number, with the highest numbers obtained under PD1 (19.1) and PD2 (18.9) conditions. The fourth and fifth plantings had significantly less leaves (18.4 and 18.5). At MAS08, both mid and late hybrids showed significantly higher total leaf numbers with the first planting, relative

Table 1
Average daily minimum (T_{min}) and maximum (T_{max}) temperatures by month for RUK07, MAS08, NGA08 and RUK08 during the 2006–2007 and 2007–2008 experiment periods.

Month	RUK07		MAS08		NGA08		RUK08	
	Max °C	Min	Max	Min	Max	Min	Max	Min
September	17.0	6.2	15.4	6.8	16.7	6.2	16.5	5.6
October	17.1	7.8	15.6	7.3	17.6	7.9	17.3	8.2
November	18.7	10.7	18.4	8.1	20.2	8.2	19.9	8.3
December	19.7	8.8	21.7	12.0	22.7	12.6	22.5	12.5
January	23.4	13.4	23.7	13.5	27.0	13.8	26.6	13.6
February	24.5	12.1	24.0	13.0	25.2	13.2	24.7	13.2
March	23.4	12.1	23.1	12.7	25.1	11.5	24.5	11.7
April	19.8	7.8	19.9	10.1	20.8	10.2	20.5	10.1
May	18.1	7.2	15.2	4.7	16.6	4.8	16.0	5.0
Average	20.2	9.6	19.7	9.8	21.3	9.8	20.9	9.8

Table 2
Mean daily radiation and rainfall totals by month for MAS08, RUK07, RUK08 and NGA08 during the 2006–2007 and 2007–2008 experiment periods.

Month	Radiation				Rainfall			
	RUK07 MJ m^{-2} d^{-1}	MAS08	NGA08[a]	RUK08	RUK07 mm	MAS08	NGA08	RUK08
September	13.9	12.8	12.8	12.8	27.6	39.8	60.4	52.2
October	15.9	15.2	18.2	18.2	125.0	62.0	97.1	97.8
November	19.2	23.2	20.7	20.7	74.8	63.2	39.4	55.6
December	22.6	21.7	21.3	21.3	67.3	44.6	74.6	64.4
January	21.2	23.9	24.4	24.4	115.1	64.0	7.2	7.0
February	21.0	22.3	20.4	20.4	37.5	46.0	35.0	40.0
March	15.2	16.6	16.4	16.4	107.8	40.0	16.9	13.8
April	11.7	11.1	11.5	11.5	32.7	94.4	133.7	133.8
May	8.0	8.1	9.3	9.3	62.0	43.6	97.8	80.0
Total	4501.6	4713.2	4717.4	4717.4	649.8	497.6	562.1	544.6

[a] Data obtained from same source as RUK08.

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Table 3
Total accumulated solar radiation and mean temperature between emergence and 50% anthesis averaged across maize hybrids for specific planting date treatments in the MAS08, RUK07, RUK08 and NGA08 environments in New Zealand, 2006–2008.

PD	Solar radiation				Mean temperature			
	RUK07 $MJ\,m^{-2}$	MAS08	NGA08	RUK08	RUK07 °C	MAS08	NGA08	RUK08
1	2130	–	1952	1971	14.4	–	15.2	15.2
2	1923	1696	1730	1788	15.3	15.7	16.2	16.3
3	1792	1585	1607	1638	16.0	16.8	17.3	17.4
4	1632	1530	1469	1507	16.9	17.8	18.7	18.8
5	1440	1593	1480	1465	17.7	18.1	19.4	19.4

to later plantings. The last two plantings had at least 1.1 less leaves than the first. Across sites, increase in mean daily temperatures (13–19 °C range) just prior to tassel initiation reduced leaf numbers by 0.1 leaf °C^{-1}.

There were significant differences in LAI among all four ENVs (Table 4), with mean values varying from 6.9 (NGA08), 6.7 (RUK08), 6.5 (RUK07) and 6.4 (MAS08). The mean air temperatures during the leaf expansion periods for the four ENVs across the PD treatments ranged from 18.6 °C (NGA08), 18.7 °C (RUK08), 17.1 °C (RUK07) and 18.0 °C (MAS08). Across all sites and hybrids there was no apparent leaf number response to photoperiod.

In general, LAI showed a quadratic response to PD, resulting in the lowest values obtained with either early or late planting. As expected, over all sites and hybrids, plants with more leaves generally had higher LAI ($r = 0.59^{***}$). Planting date had a significant impact on LAI in all ENVs except MAS08. At RUK07, PD4 had the highest LAI (7.1) while PD1 had the least LAI (6.1) (Table 4).

A significant quadratic response of LAI to PD was observed at NGA08 ($r^2 = 0.30^{**}$) and RUK08 ($r^2 = 0.33^{**}$) where LAI was maximised with PD3. In both ENVs, PD5 had the smallest LAI, followed by PD1. Despite RUK08 experiencing possible significant drought stress, the effects of water stress, particularly for PD1 to PD4, were not visibly evident during the leaf expansion phase. Precipitation received during the pre-flowering phase at RUK08 was 121, 74, 72, 69 and 47 mm for PD1 to PD5, respectively.

3.3. *Grain yield*

Across the four ENVs, the highest average GY was observed at MAS08 (16,100 kg ha^{-1}) and the lowest at RUK08 (6280 kg ha^{-1}; Table 5). Both RUK07 and MAS08 showed significant maturity × PD interactions where late hybrids significantly outyielded early hybrids when planted early and vice versa or similar with late planting (Fig. 1a and b). At RUK07 yield reduction between PD1 and PD4 was 24% and 10% for late and early hybrids, respectively. On the other hand, while yield reduction between PD4 and PD5 at MAS08 was only 7% for early hybrids, it was ≥30% for mid and late hybrids.

At RUK07, mid and late maturity hybrid yields declined linearly with PD at −45 (0.3%) and −150 kg (1%) ha^{-1} d^{-1} of planting delay. The yield response for the early group was inconsistent.

At MAS08, mid and late hybrids showed a significant quadratic yield response to PD ($r^2 = 0.96^{***}$; $r^2 = 0.83^{*}$), reaching maximum around PD3 (6 November). Like RUK07, the grain yields of early hybrids at MAS08 were more stable across all PD treatments than for later hybrids (Fig. 1a). When planting was conducted at the optimum time, early hybrids were outyielded by later hybrids.

At RUK08 and NGA08, the three maturity groups responded similarly to PD. The highest yield of 13,900 kg ha^{-1} was obtained from PD1 at NGA08. A linear response to PD delay was observed at NGA08, resulting in 35 kg ha^{-1} d^{-1} yield loss across all hybrids ($r^2 = 0.73^{***}$). Despite the very low summer rainfall (Table 2), NGA08 was only moderately affected by moisture stress due to its superior soil physical characteristics relative to RUK08. Textural differences meant that Ngaroto soils could hold approximately 119 mm of total plant available water in the top 1 m vs. 76 mm for RUK08 (Landcare Research, 2009). Maize roots at NGA08 are likely to have grown well beyond the 1 m depth, potentially tapping into additional soil water reserves. At RUK08, a sand layer existed around the 50 cm depth, reducing water holding capacity as well as root growth.

At RUK08 where moisture stress effects were more severe, yields ranged from approximately 4700 to 7300 kg ha^{-1}, with the highest yields achieved under PD1 and PD4. Despite the higher yields with PD4, a significant, but weak linear relationship constituting a yield loss of 16 kg ha^{-1} d^{-1} was observed across all hybrids ($r^2 = 0.17^{*}$). At both RUK08 and NGA08 ENVs effects of PD on yield were confounded by water stress.

Across all ENVs, significant correlations were observed between GY and: KN ($r = 0.90^{***}$); KW ($r = 0.76^{***}$); barrenness ($r = -0.72^{***}$); ears plant^{-1} ($r = 0.76^{***}$).

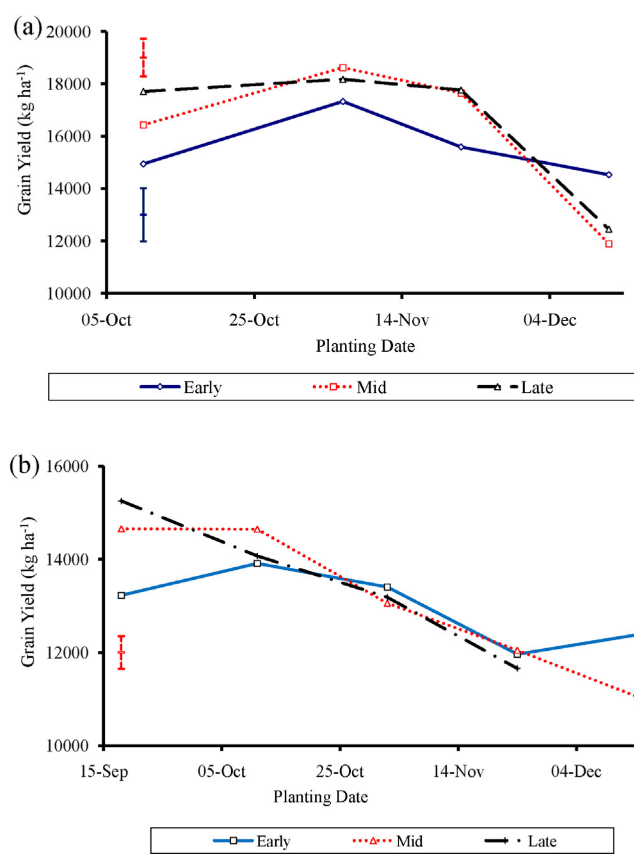

Fig. 1. Grain yields at (a) MAS08 and (b) RUK07 across 4 and 5 planting dates (PDs), respectively. The vertical bars show standard errors (se) across treatment means. The dotted bar in (a) represent se across PDs for mid and late hybrids whereas the solid bar is the se for early hybrids.

Table 4
Maize leaf area index at NGA08, RUK08, RUK07 and MAS08 environments, for 4 or 5 planting dates (PDs), averaged for 6 hybrids differing in maturity; se is standard error across all hybrids or maturities. Figures in parenthesis refer to mean temperatures (°C) during leaf expansion (between emergence and anthesis) for the respective treatment.

PD	NGA08		RUK08		RUK07		MAS08	
1	6.7	(16.9)	6.1	(16.9)	6.1	(15.8)	–	–
2	7.1	(18.1)	7.0	(18.2)	6.5	(16.1)	6.2	(17.2)
3	7.5	(18.8)	7.4	(18.8)	6.5	(16.9)	6.4	(17.9)
4	7.1	(19.4)	6.9	(19.5)	7.1	(17.7)	6.6	(18.3)
5	6.0	(19.9)	5.9	(19.9)	6.5	(18.8)	6.3	(18.7)
se	0.27		0.29		0.20		0.22	
Significance	***		***		***		NS	
Maturity								
Early	6.4		6.1		6.0		6.1	
mid	6.5		6.3		6.2		6.4	
Late	7.7		7.5		7.4		6.6	
se	0.45		0.48		0.33		0.42	
Significance	NS		NS		NS		NS	
PD × maturity								
Significance	NS		NS		NS		NS	

NS and *** denote, respectively, significance at $P >= 0.05$ and $P < 0.001$.

3.4. Biomass yield at grain harvest

Like GY, total biomass yields were highest at MAS08 and least at RUK08. At RUK07 and RUK08 the biomass yield response to PD varied by maturity (Table 6). At MAS08 a significant quadratic response of biomass yield to PD existed ($r^2 = 0.54$***), and maximum yields of 27,700 kg ha^{-1} were obtained at PD4. A linear biomass yield change of −70 kg ha^{-1} d^{-1} ($r^2 = 0.68$***) from the initial 23,700 kg ha^{-1} obtained with early planting was observed at NGA08.

At RUK07 and RUK08, total biomass for mid and late hybrids decreased linearly with PD, whereas response of early hybrids was not significant. Based on the first four plantings, mid and late hybrids recorded yield changes of about −50 kg ha^{-1} d^{-1} at RUK07. A higher rate of yield decline for late (−51 kg ha^{-1} d^{-1}) vs. mid hybrids (−37 kg ha^{-1} d^{-1}) was observed at RUK08 where biomass yields averaged 13,800 kg ha^{-1}. When compared to RUK07, drought appeared to have reduced total biomass yields for RUK08 by about 40% (Table 6).

3.5. Harvest Index and barrenness

The highest HI values were attained at MAS08 (0.53) and least at RUK08 (0.39). At RUK08 and NGA08, late hybrids had significantly lower HI values than earlier hybrids. The respective values for early, mid and late hybrids were 0.41, 0.40 and 0.36 (RUK08), 0.53, 0.51 and 0.49 (NGA08) and 0.54, 0.52 and 0.52 (MAS08).

Other than RUK07, HI response to PD was similar across hybrid maturities (Fig. 2a and b). At MAS08, a quadratic relationship between PD and HI was observed ($r^2 = 0.74$***). At NGA08 and RUK08 a significant PD effect on HI also existed while response was inconsistent at RUK07.

Barrenness was highest at RUK08 (10%) compared to 3% (RUK07), 2% (NGA08) and 0% at MAS08. For RUK08, PD2, 3 and 5 had higher barrenness levels than PD1 and 4, reflecting the timing of rainfall events rather than PD per se. Late hybrids experienced more barrenness than their earlier counterparts under stressful conditions (e.g., PD2 and PD5).

Table 5
Grain yields at 14% moisture content for NGA08, RUK08, RUK07 and MAS08 over four or five planting dates (PD); se is the pooled standard error across PD treatments for all hybrids; a and b are linear and quadratic regression estimates, expressed as kg d^{-1} from PD1.

PD	NGA08	RUK08	RUK07			MAS08		
	Hybrid maturity							
	All	All	Early	Mid	Late	Early	Mid	Late
	kg ha^{-1}							
1	13,942	6803	13,227	14,657	15,255	–	–	–
2	13,401	6423	13,912	14,648	14,073	14,937	16,427	17,707
3	13,072	6148	13,407	13,060	13,185	17,327	18,622	18,172
4	11,777	7281	11,963	12,050	11,660	15,587	17,648	17,762
5	11,128	4724	12,395	11,018	–	14527	11892	12447
se	310.7	278.4	349.5	349.5	349.5	1014.8	717.5	717.5
	Significance							
PD	***	***	***	***	***	***	***	***
Maturity	NS	NS	**	**	**	NS	NS	NS
PD × maturity	NS	NS	***	***	***	***	***	***
	Regression coefficients							
Linear	***	*	NS	***	**	NS	–	–
a	−35	−16	–	−45	−148	–	3733	2700
Quadratic	–	–	NS	–	–	NS	***	*
b	–	–	–	–	–	–	−6	−4
r^2	0.73	0.17	–	0.89	0.69	–	0.96	0.83

*, **, *** and NS denote, respectively, significance at $0.01 < P < 0.05$, $0.001 < P < 0.01$, $P < 0.001$ and $P >= 0.05$.

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Table 6
Biomass yields for NGA08, RUK08, RUK07 and MAS08 over four or five planting dates (PD); se is the pooled standard error across PD treatments for all hybrids; a and b are linear and quadratic regression estimates, expressed as kg d^{-1} from PD1.

PD	NGA08 Hybrid maturity	MAS08	RUK07			RUK08		
	All kg ha^{-1}	All	Early	Mid	Late	Early	Mid	Late
1	23,554	–	20,756	23,356	24,605	14,237	14,420	16,423
2	23,659	26,421	22,210	23,099	23,637	15,876	15,355	13,995
3	22,546	27,315	21,409	22,297	21,470	14,466	13,636	13,432
4	19,404	27,748	19,195	20,272	21,491	13,861	13,414	14,778
5	18,466	23,013	20,116	18,871	–	10,805	11,476	10,780
se	539.4	752.7	866.0	866.0	866.0	911.3	911.3	911.3
				Significance				
PD	***	***	***	***	***	***	***	***
Maturity	NS	NS	NS	NS	NS	NS	NS	NS
PD × maturity	NS	NS	***	***	***	***	***	***
				Regression coefficients				
Linear	***	**	NS	**	*	NS	**	**
a	−70	2846	–	−53	−52	–	−37	−51
Quadratic	–	**	NS	–	–	NS	NS	–
b	–	−5	–	–	–	–	–	–
r^2	0.68	0.54	–	0.71	0.59	–	0.60	0.63

*, **, *** and NS denote, respectively, significance at $0.01 < P < 0.05$, $0.001 < P < 0.01$, $P < 0.001$ and $P >= 0.05$.

Over the four ENVs, significant correlations were observed between HI and: KN ($r = 0.80$***); KW ($r = 0.64$***) and barrenness ($r = -0.78$***).

3.6. Kernel number m^{-2}

Average KN m^{-2} was highest at MAS08 and lowest at RUK08 (Table 7). NGA08 and RUK07 had comparable counts. Kernel numbers were consistent with the observed GYs in the four ENVs, reflecting the significant correlation of KN m^{-2} with GY ($r = 0.90$***). In all ENVs there was a general linear decline in KN m^{-2} with PD. The MAS08 ENV, with its optimal moisture conditions, showed a significant linear decrease of −13 kernels m^{-2} d^{-1} in planting delay. At NGA08, where drought stress was moderate, no clear PD trend existed. The lowest values were observed with PD3 suggesting that moisture stress was influencing KN. At RUK07 kernel numbers for mid and late hybrids decreased linearly with PD though the latter class responded more strongly. The observed responses were similar to those reported for GY.

3.7. Kernel weight

The highest average KW values were observed at MAS08 (0.36 g) and the lowest at RUK08 (0.16 g). Environmental differences in KW generally mimicked GY, consistent with the high correlation between GY and KW ($r = 0.76$***). At MAS08 and RUK08, KW response to PD varied depending on hybrid maturity. Though KW declined with PD at MAS08, early hybrids were more stable across PDs, varying from 0.26–0.32 g vs. 0.24–0.36 g $kernel^{-1}$ for late hybrids (Fig. 3). A quadratic response of KW to PD was observed for mid ($r^2 = 0.77$*) and late hybrids ($r^2 = 0.82$*). At RUK08, early hybrids showed greater KW variation of 0.16–0.25 g vs. mid hybrids (0.20–0.25 g). Again, early hybrids showed no definite KW trend with PD. A general decrease in KW was recorded with delays in

Table 7
Response of kernel number m^{-2} to planting date (PD) and hybrid maturity for treatments sown on 4 or 5 PDs at NGA08, RUK08, RUK07 and MAS08; a and b are linear and quadratic regression coefficients, expressed as kernels m^{-2} d^{-1} from PD1.

PD	NGA08	MAS08	RUK07			RUK08		
	Hybrid maturity							
	All Kernels m^{-2}	All	Early	Mid	Late	Early	Mid	Late
1	4526	–	4254	4025	4858	3172	2618	2944
2	4195	4787	4396	4096	4578	2796	2253	2062
3	3890	4761	4575	3844	4304	2870	2133	1975
4	3996	4675	4135	3732	4080	2621	2525	3095
5	4127	4043	4478	3615	–	2738	2142	1947
se	179.7	217.7	360.9	360.9	360.9	268.8	268.8	268.8
	Significance							
PD	***	***	***	***	***	***	***	***
Maturity	NS	NS	NS	NS	NS	NS	NS	NS
PD × maturity	NS	NS	***	***	***	***	***	***
	Regression coefficients							
Linear	NS	**	NS	*	**	NS	NS	NS
a	–	−12.9	–	−5.4	−11.7	–	–	–
Quadratic	NS	NS	NS	–	–	NS	NS	NS
b	–	–	–	–	–	–	–	–
r^2	–	0.33	–	0.50	0.75	–	–	–

*, **, *** and NS denote, respectively, significance at $0.01 < P < 0.05$, $0.001 < P < 0.01$, $P < 0.001$ and $P >= 0.05$.

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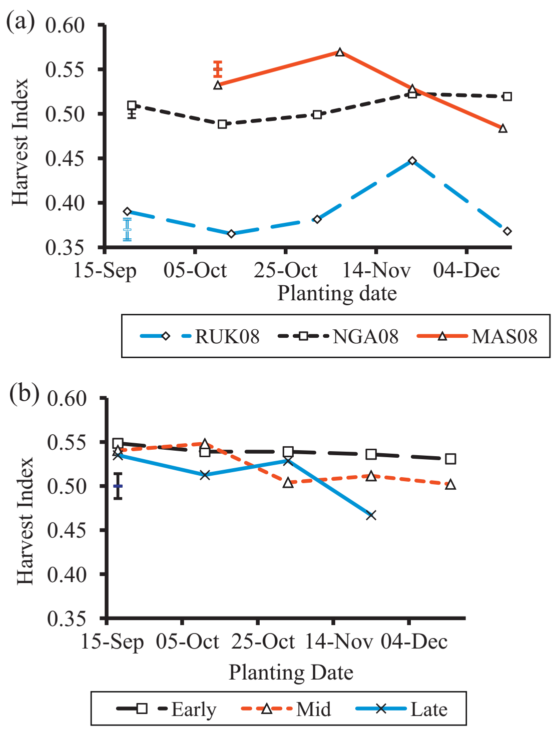

Fig. 2. Harvest index for (a) RUK08, MAS08, NGA08 and (b) RUK07 across four or five planting dates. The vertical bars show standard errors (se) across treatment means.

planting for late (-0.8 mg kernel^{-1} d^{-1}; $r^2 = 0.50^*$) and mid hybrids (-0.6 mg kernel^{-1} d^{-1}; $r^2 = 0.45^*$), but even though drought effects were not quantified, it is likely that these time trends at RUK08 were exacerbated by drought. At RUK07 and NGA08, KW was also strongly influenced by PD, resulting in a respective linear decrease of 0.5 and 0.6 mg d^{-1} from the original 0.29 or 0.28 g kernel^{-1}.

4. Discussion

4.1. Leaf area index

Leaf area index for late hybrids in Waikato was, on average, about 20% higher than that of early and mid hybrids. Even though yield differences between early and late hybrids are largely attributed to the longer period of radiation interception for the latter (Tsimba et al., 2013), at the individual plant level, an increase in the LA of later hybrids could confer an advantage. The larger LA per plant for later maturing hybrids can be expected to increase

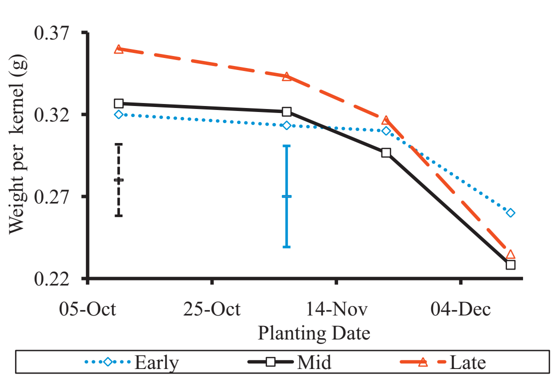

Fig. 3. Weight per kernel at MAS08 across 4 planting dates (PDs). The vertical bars show standard errors (se) across treatment means. The solid bar is se across PDs for early hybrids whereas the dotted bar is the pooled se across PDs for mid and late hybrids.

IPAR per plant. Differences in LAI observed here confirm the need to increase planting density for early maturing hybrids in order to to match photosynthetically active radiation interception of later hybrids.

The observed differences in LAI among ENVs were largely due to temperature variations though inclusion of earlier hybrids in the Manawatu ENV was largely responsible for the lower LAI values at MAS08. Even under drought conditions, water stress effects appear to have occurred after LA expansion was complete. Wilson et al. (1973) showed that temperatures <18 °C and >25 °C lowered LAI. Temperature regimes of about 15–17 °C under early planting conditions in the current study thus reduced LAI. Lower leaf numbers in some ENVs could also have slightly contributed to lower LAI under very late planting conditions.

4.2. Grain yield, yield components and grain traits

The general quadratic response in yield to PD for mid and late hybrids under the higher latitude conditions of Manawatu, where spring temperatures rose more slowly, was an indication that planting too early decreased yields through a source limitation attributed to low temperatures and reduced canopy development.

In a normal season (2006/07), late hybrids significantly outyielded earlier hybrids when planted early and vice versa under delayed planting conditions. Lower yields in early hybrids (vs. late), particularly when sown early, are usually due to source limitation caused by smaller LA, lower daily and seasonal IPAR and cooler temperatures (Stone et al., 1999). The smaller ear and sink size of early hybrids also meant that under optimum conditions, early hybrids were sink-limited. In Waikato, flowering dates of all hybrids, particularly the early group, coincided with the highest irradiance levels just after 21 December (summer solstice), which should have resulted in the greatest assimilate flux to the developing ear. Due to their larger LA, bigger sink size and longer grain filling duration, late vs. early hybrids would have accumulated more IPAR plant^{-1}, resulting in higher rates of DM accumulation over a longer period.

Across all ENVs, early maturing hybrids yielded consistently at all PDs, suggesting that they had an appropriate source–sink balance for stable yields, but failed to fully exploit the advantages of early planting. Short duration hybrids should therefore be planted at higher densities to compensate for their lower LA per plant and their inherently smaller ear size if they are to be sown early. Long duration hybrids however, responded to a longer growing season with greater GY when sown early. When late planted, longer duration hybrids may be more prone to autumn frosts, especially in high risk areas. They should therefore be replaced by their earlier counterparts to minimise frost risk in autumn and also reduce grain drying costs. Hybrid choice and planting density should therefore be based on potential to fully utilise available resources for the duration of the growing ENV.

With late planting, lower GYs were largely due to the grain filling phase coinciding with diminishing temperature and radiation levels and hence, a decline in KW (Fig. 3). Autumn temperature and IPAR conditions tended to deteriorate more rapidly in the higher latitude ENV, reducing RUE (Andrade et al., 1993; Wilson et al., 1995) and slowing grain growth (Ruget, 1993). For instance, the higher latitude ENV average radiation and mean temperatures during the second half of PD5 grain filling were 11 MJ m^{-2} d^{-1} and 14.8 °C, vs. 20 MJ m^{-2} d^{-1} and 18.1 °C (PD1).

The decline in GY associated with late planting was greater for late than early hybrids. The longer grain filling duration of late hybrids (Tsimba et al., 2013) can increase their exposure to adverse conditions such as drought which may amplify the yield decline. For these reasons KN and KW were also more stable in early than in later hybrids.

In ENVs where water stress is likely to occur later in the season, yield variation is likely to be influenced more by water stress than PD. Early plantings would either have completed a significant part of their life cycle and/or established a deep root system before water stress becomes severe, allowing them to access residual moisture from depth.

Soil type, and in particular, soil water holding capacity and rooting depth, will have a significant influence on drought effects as noted by the contrast between RUK08 and NGA08. For instance, using data from PD1 and PD2, volumetric water contents at RUK08 (top 1 m; data not shown) varied significantly during the season, falling from 26% (28 February) to 12% (17 March), vs. a drop from 23% to 21% at NGA08 over the same period. Considering biomass production at each site, textural and depth differences together, plants growing on NGA08 soils were accessing approximately twice the volume of stored soil water than at RUK08, suggesting about a two-fold difference in effective rooting depth between these sites.

The adjustment of PD to synchronise water stress sensitive stages with periods of adequate moisture levels cannot be foreseen at planting time. Consequently, on drought prone soils it may be strategic to plant two hybrids differing in maturity on the same date to reduce risk of drought stress at flowering. In drought situations there were generally no advantages in planting late hybrids other than for risk management.

The strong correlation between KN m^{-2} and GY (r = 0.90***) agrees with other reports in the literature (e.g., Otegui et al., 1995; Bolaños and Edmeades, 1996). The higher correlation between GY and KN vs. KW was expected since kernel weight is more strongly conserved than KN in response to varying environmental conditions (Westgate et al., 1997; Borrás et al., 2009). This was further evidenced by the general stability of KW across PDs at the drought stressed site (RUK08). Gambín et al. (2008) also reported that while enhanced plant growth rate increased KW in sorghum, this was less important in maize since maximum kernel size was defined primarily during the lag phase of grain filling.

Kernel number per plant usually reflects assimilate availability two weeks either side of flowering. At RUK08, the lowest KNs were obtained when either rainfall or irradiance levels around flowering were low. For instance, PD3 which received only 3.8 mm rainfall and averaged 25 MJ m^{-2} d^{-1} solar radiation 2 wk either side of flowering, and PD5 which received 23 mm rainfall but lower irradiance (18 MJ m^{-2} d^{-1}; data not shown), had the lowest KN. On the other hand, PD4 and PD1 with rainfall >20 mm and radiation >24 MJ m^{-2} d^{-1} around flowering, had the highest KNs. Increased KN thus corresponded with greater flux of assimilate to the ear at flowering time (Kiniry and Knievel, 1995; Andrade et al., 2000).

Drought and late planting generated similar changes in GY and yield components of early hybrids, suggesting that strategies that mitigate the limited source and sink capacities of early hybrids should help reduce effects of both stresses and enhance yields under optimum conditions. For example, sink limitation due to small ears in early hybrids (Westgate et al., 1997; Sarlangue et al., 2007) can be countered by increasing planting densities.

4.3. Total aboveground biomass and harvest index

In general, total aboveground biomass responded to PD and ENV in a similar manner to GY. This was expected since the two are not independent of each other. The decrease in biomass with PD in later maturing hybrids reflected the greater impact of late planting on long duration hybrids, especially under conditions of diminishing IPAR and temperature. As with GY, total biomass for early hybrids was stable across PDs, a result that could also be attributed to sink limitation or an appropriate balance between the supply of environmental factors (source) for growth and crop duration.

The quadratic response of total biomass to PD across most ENVs was largely due to low temperatures during early growth which decreased LA in early PD treatments. Low late season radiation and temperature levels under delayed planting regimes also decreased biomass yields through reductions in IPAR and RUE.

Grain formation, because it occurs in the last 60% of the plant's life, was more affected by late planting than total biomass production, resulting in a decrease in HI. In Waikato, the decline in HI with PD was greater for late than early hybrids and this was most likely due to late hybrids being exposed to low temperature and IPAR for a greater proportion of their life cycles. In the lower latitude ENVs where autumn temperatures did not fall as quickly, the decline in HI with PD observed in late hybrids was correspondingly less. Early hybrids, because of their shorter duration and smaller LA per plant, produced less biomass than late hybrids, resulting in stable yields of grain and biomass. The smaller sink size of early hybrids also meant that assimilate demand could be more easily met promoting more complete grain fill and thus, more stable HI values under late plantings. The higher correlation between HI and KN vs. other yield components indicates that KN was more important than KW in driving the decline in HI.

5. Conclusions

Early plantings experienced low early season temperatures which reduced biomass and LA expansion but increased leaf number. Conversely, late plantings experienced higher summer temperatures that reduced leaf numbers and time to reach silking, consequently lowering photothermal quotients (the ratio of IPAR to thermal time).

Under higher latitude conditions maize yields responded in a quadratic manner to PD with maximum yields corresponding to PDs in mid to late October. This was attributed to cool early season temperatures followed by diminishing temperatures (<15 °C) and solar radiation (<11 MJ m^{-2} d^{-1}) during mid to late autumn. In the lower latitude ENV, these effects were weaker because spring temperatures warmed more quickly while radiation and temperature declined more slowly in autumn.

Late planting resulted in a significant deterioration of environmental conditions (temperature, radiation and water availability) during grain filling, suggesting a caution on delayed planting in drought prone soils, at higher elevations, or in cooler and shorter season environments. Late maturing hybrids when sown late in the season were generally more susceptible to any form of stress that increased in intensity with time (e.g., drought, declining temperatures) when compared to early hybrids.

The effects of extreme drought masked PD response, favouring early maturing hybrids compared to late hybrids. This management x environment interaction highlights a major challenge encountered when agronomic field experiments are being used to make crop management decisions. For this reason, crop simulation models linked to long-term weather databases are a very useful option.

An important future research priority would be to reassess the optimal planting density for short duration hybrids when they are being planted early. Sink and source limitation in early hybrids could be mitigated by modest increases in planting densities without compromising stability of grain and biomass yields. Early plantings are recommended in ENVs likely to dry out in summer to allow roots to develop to depth before soil water reserves become limiting.

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References

Al-Darby, A.M., Lowery, B., 1987. Seed zone temperature and early corn growth with three conservation tillage systems. Proc. Soil Sci. Soc. Am. 28, 272–275.

Andrade, F.H., Uhart, S.A., Cirilo, A., 1993. Temperature affects radiation use efficiency in maize. Field Crops Res. 32, 17–25.

Andrade, F.H., Otegui, M.E., Vega, C., 2000. Intercepted radiation at flowering and kernel number in maize. Agron. J. 92, 92–97.

Bolaños, J., Edmeades, G.O., 1996. The importance of the anthesis-silking interval in breeding for drought tolerance in tropical maize. Field Crops Res. 48, 65–80.

Borrás, L., Zinselmeier, C., Senior, M.L., Westgate, M.E., Muszynski, M.G., 2009. Characterization of grain-filling patterns in diverse maize germplasm. Crop Sci. 49, 999–1009.

Edmeades, D.C., Wheeler, D.M., Waller, J.E., 1984. Comparison of methods for determining lime requirements of New Zealand soils. N. Z. J. Agric. Res. 28, 93–100.

Edmeades, G.O., Bolaños, J., Elings, A., Ribaut, J.-M., Bänziger, M., Westgate, M.E., 2000. The role and regulation of the anthesis-silking interval in maize. In: Westgate, M.E., Boote, K. (Eds.), Physiology and Modeling Kernel Set in Maize. CSSA, Madison, WI, USA, pp. 43–73.

Elings, A., 2000. Estimation of leaf area in tropical maize. Agron. J. 92, 436–444.

Gambín, B.L., Borrás, L., Otegui, M.E., 2008. Kernel weight dependence upon plant growth at different grain-filling stages in maize and sorghum. Aust. J. Agric. Res. 59, 280–290.

Gomez, K.A., Gomez, A.A., 1984. Statistical Procedures for Agricultural Research, second ed. John Wiley and Sons, Singapore.

Hewitt, A.E., 1998. New Zealand Soil Classification. Landcare Research Science Series No. 1, 2nd ed. Manaaki Whenua Press, Lincoln, New Zealand.

Johnson, R.R., Mulvaney, D.L., 1980. Development of a model for use in maize replant decisions. Agron. J. 72, 459–464.

Jones, J.W., Zur, B., Bennett, J.M., 1986. Interactive effects of water and nitrogen stresses on carbon and water vapor exchange of corn canopies. Agric. Forest Meteorol. 38, 113–126.

Kiniry, J.R., Knievel, D.P., 1995. Response of maize seed number to solar radiation intercepted soon after anthesis. Agron. J. 87, 228–234.

Lammi, J., 2008. Photoperiod calculator: Computation of daylengths, sunrise/sunset times, twilight and local noon. http://www.martindalecenter.com/CalculatorsD_Wea.html (verified on 24.11.09).

Landcare Research, 2009. Soil Report. http://www.smap.landcare.co.nz (verified on 29.08.09).

Lauer, J.G., 1998. The Wisconsin Comparative Relative Maturity (CRM) System for Corn. http://www. corn.agronomy.wisc.edu/AA/A021.aspx (verified on 8.03.09).

Littell, R.C., Milliken, G.A., Stroup, W.W., Wolfinger, R.D., Schabenberger, O., 2006. SAS for Fixed Models, 2nd ed. SAS Institute Inc., Cary, NC.

Milliken, G.A., Johnson, D.E., 1994. Analysis of Messy Data. Volume 1. Designed Experiments, 2nd ed. Chapman & Hall, London.

Monteith, J.L., 1977. Climate and the efficiency of crop production in Britain. Philos. Trans. R. Soc. Ser. B 281, 277–329.

Muchow, R.C., Carberry, P.S., 1989. Environmental control of phenology and leaf growth in tropical adapted maize. Field Crops Res. 20, 221–236.

NIWA Science, 2001. Overview of New Zealand climate. http://www.niwascience.co.nz (verified on 4.02.09).

Otegui, M.E., Andrade, F.H., Suero, E.E., 1995. Growth, water use, and kernel abortion of maize subjected to drought at silking. Field Crops Res. 40,87–94.

Otegui, M.E., Ruiz, R.A., Petruzzi, D., 1996. Modeling hybrid and sowing date effects on potential grain yield of maize in a humid temperate region. Field Crops Res. 47, 167–174.

Ruget, F., 1993. Contribution of storage reserves during grain filling of maize in northern European conditions. Maydica 38, 51–59.

Sarlangue, T., Andrade, F.H., Calviño, P., Purcell, L., 2007. Why do maize hybrids respond differently to variations in plant density? Agron. J. 99, 984–991.

Sorensen, I., Stone, P., Rogers, B., 2000. Effect of sowing time on yield of a short and a long season maize hybrid. Proc. Agron. Soc. N.Z. 30, 63–66.

Staggenborg, S.A., Fjell, D.L., Devlin, D.L., Gordon, W.B., Maddux, L.D., Marsh, B.H., 1999. Selecting optimum planting dates and plant populations for dryland corn in Kansas. J. Prod. Agric. 12, 85–90.

Stone, P.J., Sorensen, I., Jamieson, P.D., 1999. Effect of soil temperature on phenology, canopy development, biomass and yield of maize in a cool-temperate climate. Field Crops Res. 63, 169–178.

Tsimba, R., 2011. Development of a decision support system to determine the best maize (Zea mays L.) hybrid-planting date option under typical New Zealand management systems. Massey University, New Zealand, pp. 261 (Ph.D. Thesis).

Tsimba, R., Edmeades, G.O., Millner, J.P., Kemp, P.D., 2013. The effect of planting date on maize: Phenology, thermal time durations and growth rates in a cool temperate climate. Field Crops Res., http://dx.doi.org/10.1016/j.fcr.2013.05.021.

Westgate, M.E., Forcella, F., Reicosky, D.C., Somsen, J., 1997. Rapid canopy closure for maize production in the northern US Corn Belt: Radiation-use efficiency and grain yield. Field Crops Res. 49, 249–258.

Wilson, D.R., Muchow, R.C., Murgatroyd, C.J., 1995. Model analysis of temperature and solar radiation limitations to maize potential productivity in a cool climate. Field Crops Res. 43, 1–18.

Wilson, J.H., Clowes, M.S.J., Allison, J.C.S., 1973. Growth and yield of maize at different altitudes in Rhodesia. Ann. Appl. Biol. 73, 77–84.